

# Suffering 1

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## 1 If $r$ is rational ( $r \neq 0$ ) and $x$ is irrational, prove that $r + x$ and $rx$ are irrational.

Since  $r$  is rational, there exist integers  $p$  and  $q > 0$  such that  $r = \frac{a}{b}$ .

### 1.1 $r + x$

Let us first show  $r + x$  is irrational. We will prove this using contradiction: Assume  $r + x$  is rational, and there exist integers  $c$  and  $d > 0$  such that  $r + x = \frac{c}{d}$ .

Since we know  $r = \frac{a}{b}$ , we can substitute  $\frac{a}{b}$  for  $r$ . We can then isolate  $x$  to get

$$\begin{aligned}x &= \frac{c}{d} - \frac{a}{b} \\x &= \frac{bc - ad}{bd}\end{aligned}$$

Since the integers are closed under addition and multiplication, it must be the case that  $bc - ad$  is an integer. Furthermore, since  $b > 0$  and  $d > 0$ , it must be the case that  $bd$  is an integer greater than 0. This is a contradiction, as now there exist integers  $p = bc - ad$  and  $q = bd > 0$  such that  $x = \frac{p}{q}$  (meaning  $x$  would be rational), but this cannot be true since  $x$  is irrational.

### 1.2 $rx$

We can prove  $rx$  is irrational in the same way. Once again, we assume  $rx$  is rational and  $rx = \frac{c}{d}$  for integers  $c$  and  $d > 0$ .

We once again substitute  $\frac{a}{b}$  for  $r$  and isolate  $x$ :

$$\frac{a}{b}x = \frac{c}{d}$$

$$x = \frac{bc}{ad}$$

Since integers are closed under multiplication,  $bc$  and  $ad$  are integers, with  $ad > 0$  because  $a > 0$  and  $d > 0$ . Thus, since  $x$  can be expressed as  $\frac{p}{q}$  for integers  $p$  and  $q > 0$ , we derive the contradiction that  $x$  is rational.

## 2 Prove that there is no rational number whose square is 12.

We will prove this using contradiction. Let  $x$  be a rational number whose square is 12. This means  $x = \frac{a}{b}$  for integers  $a$  and  $b > 0$ ; where  $a$  and  $b$  are also coprime.

We will start by assuming  $x^2 = 12$ .

$$x^2 = 12$$

$$\frac{a^2}{b^2} = 12$$

$$a^2 = 12b^2$$

From this, we can see that  $a^2$  is a multiple of 3, as  $12b^2$  is a multiple of 3. Since  $a^2$  is a multiple of 3,  $a$  is a multiple of 3. As such, we can actually conclude that  $a^2$  is a multiple of 9 knowing that  $a$  is a multiple of 3.

This would lead us to believe that  $12b^2$  is also a multiple of 9, since  $a^2$  is. However, this would require  $b$  to be a multiple of 3, which cannot be true, as  $a$  and  $b$  are coprime and  $a$  is already a multiple of 3. As such, by contradiction, we prove that there is no rational  $x$  whose square is 12.

## 3 Prove Proposition 1.15

### 3.1 a) If $x \neq 0$ and $xy = xz$ then $y = z$

We follow the same strategy as the proof for Proposition 1.14.

We let  $y = 1 \cdot y$ . Furthermore, we know  $x^{-1} \cdot x = 1$  since  $x \neq 0$ . Thus, we can say

$$y = x^{-1} \cdot x \cdot y$$

$$y = x^{-1}(xy)$$

$$y = x^{-1}(xz)$$

$$y = x^{-1} \cdot x \cdot z$$

$$y = z$$

**3.2 b) If  $x \neq 0$  and  $xy = x$  then  $y = 1$**

Since  $x \cdot 1 = x$ , we know  $xy = x \cdot 1$ . Since we proved part a and  $x \neq 0$ , we can use that statement to show  $y = 1$  if we let  $z = 1$ .

**3.3 c) If  $x \neq 0$  and  $xy = 1$  then  $y = x^{-1}$**

We know  $x \cdot x^{-1} = 1$ , so we can see that  $xy = x \cdot x^{-1}$ . Once again, since  $x \neq 0$  and  $xy = x \cdot x^{-1}$ , we have  $y = x^{-1}$  from part a.

**3.4 d) If  $x \neq 0$  then  $(x^{-1})^{-1} = x$**

We know that  $x^{-1} \cdot (x^{-1})^{-1} = 1$ , and  $x \cdot x^{-1} = 1$ . Thus,  $x^{-1} \cdot (x^{-1})^{-1} = x^{-1} \cdot x$ , and since  $x \neq 0$  part a gives us  $(x^{-1})^{-1} = x$ .

**4 Let  $E$  be a nonempty subset of an ordered set; suppose  $\alpha$  is a lower bound of  $E$  and  $\beta$  is an upper bound of  $E$ . Prove that  $\alpha \leq \beta$ .**

Let  $x$  be an arbitrary element in  $E$ . Since  $\alpha$  is a lower bound of  $E$ ,  $\alpha \leq x$ . Since  $\beta$  is an upper bound of  $E$ ,  $\beta \geq x$ . Thus,  $\alpha \leq x \leq \beta$ , so  $\alpha \leq \beta$ .

**5 Let  $A$  be a nonempty set of real numbers which is bounded below. Let  $-A$  be the set of all numbers  $-x$ , where  $x \in A$ . Prove that**

$$\inf A = -\sup(-A)$$

Let  $y = \inf A$ . Then, it must be the case that for all  $x \in A$ ,  $y \leq x$ . Let's try to show that  $-y$  is also an upper bound of  $-A$ . Consider any  $-x \in -A$ . This means  $x \in A$ , and we already know that  $y \leq x$ . Thus,  $-y \geq -x$ , so for any  $-x \in -A$ ,  $-y \geq -x$ .

We must now show that  $-y$  is also the least upper bound of  $-A$ . Since  $y$  is the greatest lower bound of  $A$ , we know that, for all  $z > y$ ,  $z$  is not a lower bound of  $A$ . This means that there exists  $x \in A$  such that  $x < z$ .

Consider any  $-z < -y$ . We want to show that for all  $-z < -y$ ,  $-z$  is not an upper bound of  $-A$ . This would make  $-y$  the least upper bound of  $-A$ . Since  $-z < -y$ , it is the case that  $z > y$ .

Furthermore, we already know that for any  $z > y$ , there exists  $x \in A$  such that  $x < z$  ( $z$  is not a lower bound of  $A$ ). Thus, there also exists  $-x \in -A$  such that  $-x > -z$ , making  $-z$  not an upper bound of  $-A$ .

We can now conclude that  $-y = \sup(-A)$ . Thus, we have  $y = -\sup(-A)$ , as well as  $y = \inf A$ , so  $\inf A = -\sup(-A)$ .

## 6 Fix $b > 1$ .

**6.1 a. If  $m, n, p, q$  are integers,  $n > 0, q > 0$ , and  $r = \frac{m}{n} = \frac{p}{q}$ , prove that**

$$(b^m)^{\frac{1}{n}} = (b^p)^{\frac{1}{q}}.$$

**Hence it makes sense to define  $b^r = (b^m)^{\frac{1}{n}}$ .**

Since  $\frac{m}{n} = \frac{p}{q}$ , we know  $mq = np$ . Thus, we also have:

$$b^{mq} = b^{np}$$

Since  $m, n, p, q$  are integers,

$$b^{mq} = (b^m)^q \text{ and } b^{np} = (b^n)^p$$

This is because we can expand  $x^{ab}$  into  $x \times x \times \cdots \times x$   $ab$  times, which can be then grouped into  $(x^a) \times (x^a) \times \cdots \times (x^a)$   $b$  times, which is equal to  $(x^a)^b$ .

$$(b^m)^q = (b^p)^n$$

$$b^m = ((b^p)^n)^{\frac{1}{q}}$$

$$(b^m)^{\frac{1}{n}} = (((b^p)^n)^{\frac{1}{q}})^{\frac{1}{n}}$$

Here, we know intuitively that the  $n$  and  $\frac{1}{n}$  exponents should cancel. Thus, we should try to work the RHS of the equation so that we can do so. Let's show that for integers  $n, q > 0$ ,  $(x^{\frac{1}{n}})^{\frac{1}{q}} = (x^{\frac{1}{q}})^{\frac{1}{n}}$  for any real number  $x$ .

**Lemma 6.1** *If  $x > 0$  is a real number and  $n, q$  are integers  $> 0$ , then  $(x^{\frac{1}{n}})^{\frac{1}{q}} = (x^{\frac{1}{q}})^{\frac{1}{n}}$*

To prove this, we raise both of the terms we're comparing to the  $nq$ -th power.

$$\begin{aligned} ((x^{\frac{1}{n}})^{\frac{1}{q}})^{nq} &= (((x^{\frac{1}{n}})^{\frac{1}{q}})^q)^n = (x^{\frac{1}{n}})^n = x \\ ((x^{\frac{1}{q}})^{\frac{1}{n}})^{nq} &= (((x^{\frac{1}{q}})^{\frac{1}{n}})^n)^q = (x^{\frac{1}{q}})^q = x \end{aligned}$$

By Theorem 1.21 in the textbook, for every real  $x > 0$  and every integer  $n > 0$  there one unique  $y$  such that  $y^n = x$ . Thus, since  $nq$  is an integer, and both  $((x^{\frac{1}{n}})^{\frac{1}{q}})^{nq}$  and  $((x^{\frac{1}{q}})^{\frac{1}{n}})^{nq}$  equal  $x$ , it must be the case that  $(x^{\frac{1}{n}})^{\frac{1}{q}} = (x^{\frac{1}{q}})^{\frac{1}{n}}$ .

We return to where we left off:

$$(b^m)^{\frac{1}{n}} = (((b^p)^n)^{\frac{1}{q}})^{\frac{1}{n}}$$

By Lemma 6.1,

$$\begin{aligned} (b^m)^{\frac{1}{n}} &= (((b^p)^n)^{\frac{1}{n}})^{\frac{1}{q}} \\ (b^m)^{\frac{1}{n}} &= (b^p)^{\frac{1}{q}} \end{aligned}$$

We have achieved the desired result.

## 6.2 b. Prove that $b^{r+s} = b^r \cdot b^s$ if $r$ and $s$ are rational.

As with many other proofs with rationals, we let  $r = \frac{m}{n}$  for some integers  $m, n$  and  $n > 0$ , and  $s = \frac{p}{q}$  for some integers  $p, q$  and  $q > 0$ .

Then, we have:

$$\begin{aligned} b^{r+s} &= b^{\frac{m}{n} + \frac{p}{q}} = b^{\frac{mq+np}{nq}} \\ b^r \cdot b^s &= b^{\frac{m}{n}} \cdot b^{\frac{p}{q}} \end{aligned}$$

We will use a similar strategy to the previous part. Since  $b > 0$ , both  $b^{r+s}$  and  $b^r \cdot b^s$  are positive. Thus, if  $(b^{r+s})^{nq} = (b^r \cdot b^s)^{nq}$ , then it means  $b^{r+s} = b^r \cdot b^s$  by Theorem 1.21.

$$(b^{r+s})^{nq} = (b^{\frac{mq+np}{nq}})^{nq}$$

By part a,  $b^{\frac{mq+np}{nq}} = (b^{mq+np})^{\frac{1}{nq}}$ .

$$\begin{aligned} ((b^{mq+np})^{\frac{1}{nq}})^{nq} &= b^{mq+np} \\ (b^r \cdot b^s)^{nq} &= (b^{\frac{m}{n}} \cdot b^{\frac{p}{q}})^{nq} \end{aligned}$$

Since  $nq$  is a positive integer, we can distribute the  $nq$  into the inner expression, as we can expand and rearrange the repeated multiplication into

$$\begin{aligned} & (b^{\frac{m}{n}})^{nq} \cdot (b^{\frac{p}{q}})^{nq} \\ &= (((b^m)^{\frac{1}{n}})^n)^q \cdot (((b^p)^{\frac{1}{q}})^q)^n \\ &= b^{mq} \cdot b^{np} \\ &= b^{mq+np} \end{aligned}$$

We have shown that  $(b^{r+s})^{nq} = (b^r \cdot b^s)^{nq}$ , so it must be the case that  $b^{r+s} = b^r \cdot b^s$ .

**6.3 c. If  $x$  is real, define  $B(x)$  to be the set of all numbers  $b^t$ , where  $t$  is rational and  $t \leq x$ . Prove that**

$$b^r = \sup B(r)$$

**when  $r$  is rational. Hence it makes sense to define**

$$b^x = \sup B(x)$$

**for all real  $x$ .**

To prove that  $b^r = \sup B(r)$ , we must show two things:

First, we must show that for all  $b^y \in B(r)$ ,  $b^r \geq b^y$ . Since  $r$  and  $y$  are rational, let  $r = \frac{m}{n}$  where  $m, n$  are integers and  $n > 0$ , and let  $y = \frac{p}{q}$  where  $p$  and  $q$  are integers and  $q > 0$ .

Since  $b^y \in B(r)$ , we have  $y \leq r$ . If we can show  $\frac{b^r}{b^y} \geq 1$ , then we know  $b^r \geq b^y$ . Let's substitute  $r$  and  $y$  with their respective fractions.

$$\begin{aligned} & \frac{b^{\frac{m}{n}}}{b^{\frac{p}{q}}} \\ &= b^{\frac{m}{n}} \cdot b^{-\frac{p}{q}} \end{aligned}$$

By part b, we have

$$\begin{aligned} & b^{\frac{m}{n} - \frac{p}{q}} \\ &= b^{\frac{mq - np}{nq}} \end{aligned}$$

We would like to show that this value is larger than or equal to 1.

**Lemma 6.2** *If  $b > 1$  is a real number, then  $b^x \geq 1$  if  $x$  is a rational number and  $x \geq 0$ .*

Assume  $x > 0$ , since  $x = 0$  implies  $b^x = 1$ , which satisfies  $b^x \geq 1$ .

Since  $x$  is rational, we write  $x$  as  $\frac{c}{d}$ , where  $c, d$  are integers and  $c > 0$  and  $d > 0$ , since  $x > 0$ .

Now,  $b^x = b^{\frac{c}{d}} = (b^c)^{\frac{1}{d}}$  from part a. Now, let's raise this term to the  $d$ -th power.

$$\frac{((b^c)^{\frac{1}{d}})^d}{b^c}$$

Since  $c > 1$ ,  $b^c$  is repeated multiplication of  $b$ , which must result in a number greater than 1. We will prove this inductively using Proposition 1.18 part b.

Our base case is  $c = 1$ , and we have  $b^c = b^1 = b > 1$ .

Assume  $b^{c-1} > 1$ . Since  $b^{c-1} > 1$  and  $b > 0$  and  $b^c = b^{c-1} \cdot b$ , Proposition 1.18 b tells us that if  $b^{c-1} > 0$ , and  $1 < b$ , then  $b^{c-1} \cdot 1 < b^{c-1} \cdot b = b^c$ , so  $b^c > b^{c-1} > 1$ .

We have shown that  $b^{\frac{mq-np}{nq}} \geq 1$ , which means  $b^r \geq b^y$ . Since  $b^y$  is an arbitrary member of  $B(r)$ ,  $b^r$  is an upper bound of  $B(r)$ .

We must also show that  $b^r$  is the least upper bound. Let  $z < b^r$ . Since  $z < b^r$  and  $b^r \in B(r)$ , clearly  $z$  is not an upper bound of  $B(r)$ . We have now shown that  $b^r$  is the least upper bound of  $B(r)$ , so  $b^r = \sup B(r)$ .

#### 6.4 d. Prove that $b^{x+y} = b^x \cdot b^y$ for all real $x$ and $y$ .

We will try to prove this by showing  $b^{x+y} \geq b^x \cdot b^y$  and  $b^{x+y} \leq b^x \cdot b^y$ .

Assume for sake of contradiction that  $b^{x+y} < b^x \cdot b^y$ . This gives

$$b^x > \frac{b^{x+y}}{b^y}$$

Since  $b^x = B(x) > \frac{b^{x+y}}{b^y}$ , there exists some rational  $s$  such that  $b^s > \frac{b^{x+y}}{b^y}$ . We then get

$$b^s \cdot b^y > b^{x+y}$$

We repeat the process with  $y$ .

$$b^y > \frac{b^{x+y}}{b^s}$$

By similar logic, there exists  $b^t$  with  $t \leq y$  and  $t$  rational such that

$$\begin{aligned} b^t &> \frac{b^{x+y}}{b^s} \\ b^s \cdot b^t &> b^{x+y} \end{aligned}$$

However, since  $b^s \cdot b^t = b^{s+t}$  since  $s, t \in \mathbb{Q}$ ,

$$b^{s+t} > b^{x+y}$$

is a contradiction since  $s + t \leq x + y$ . Thus, it is the case that  $b^{x+y} \geq b^x \cdot b^y$ . We must now show that  $b^{x+y} \leq b^x \cdot b^y$ .

To do this, we introduce the idea of  $B'(x + y)$ , which is the set of all  $b^t$  such that  $t \in \mathbb{Q}$  and  $t < x + y$ . We first show that  $b^{x+y} = \sup B(x + y) = \sup B'(x + y)$ .

It should be clear that  $\sup B(x + y)$  is an upper bound of  $B'(x + y)$ , since  $B'(x + y)$  is a subset of  $B(x + y)$ . We then need to show that  $\sup B'(x + y) \geq \sup B(x + y)$ .

We will prove this through contradiction. Assume there exists some  $t \leq x + y$  such that  $b^t > \sup B'(x + y)$ . This means  $\frac{b^t}{\sup B'(x + y)} > 1$ . From problem 7 part c, since  $\frac{b^t}{\sup B'(x + y)} > 1$ , if a positive integer  $n > \frac{b^t}{\frac{b^t}{\sup B'(x + y)}}$ , then  $b^{\frac{1}{n}} < \frac{b^t}{\sup B'(x + y)}$ . Of course  $n$  must exist. Thus,

$$b^{\frac{1}{n}} < \frac{b^t}{\sup B'(x + y)}$$

This then implies

$$b^{\frac{1}{n}} \cdot \sup B'(x + y) < b^t$$

$$\sup B'(x + y) < \frac{b^t}{b^{\frac{1}{n}}} = b^{t - \frac{1}{n}}$$

This is a contradiction, as  $t - \frac{1}{n}$  is a rational number less than  $x + y$ . This means  $b^{t - \frac{1}{n}} \in B'(x + y)$ , so it cannot be possible that  $\sup B'(x + y)$  is an upper bound of  $B'(x + y)$ .

With this, we have shown that  $\sup B(x + y) = b^{x+y} = \sup B'(x + y)$ . Now it is our goal to show that  $b^x \cdot b^y \geq \sup B'(x + y) = b^{x+y}$ .

Consider any arbitrary element  $b^r \in B'(x + y)$ . Here,  $r$  is a rational number less than  $x + y$ . Let  $0 < \epsilon = x + y - r$ .

Now, we can let  $s$  be a rational number such that  $x - \epsilon < s < x$ . Since rationals are dense in the reals,  $s$  exists. Then, we can also define  $t$  to be  $r - s$ . Because we have

$$x - \epsilon < s < x$$

If we multiply the inequality by  $-1$  we get

$$\epsilon - x > -s > -x$$



Adding  $r$  to all expressions,

$$r + \epsilon - x > r - s > r - x$$

Recall that  $\epsilon = x + y - r$

$$r + x + y - r - x > r - s > r - x$$

$$y > r - s > r - x$$

Since  $r - s = t$ ,

$$y > t > r - x$$

Thus,  $y > t$  and  $x > s$ , while  $s + t = r$ . Because  $x > s$ ,  $b^x > b^s$ . Similarly,  $y > t$  implies  $b^y > b^t$ . Thus,

$$b^x \cdot b^y > b^s \cdot b^t = b^{s+t} = b^r$$

We can then conclude that

$$b^x \cdot b^y > b^r$$

for all rational  $r < x + y$ . Therefore,  $b^x \cdot b^y$  is an upper bound of  $B'(x + y)$ , meaning  $b^x \cdot b^y \geq \sup B'(x + y) = b^{x+y}$ .

We have proven that  $b^x \cdot b^y = b^{x+y}$  since we proved that both  $b^{x+y} \geq b^x \cdot b^y$  and  $b^x \cdot b^y \geq b^{x+y}$  are true.

## 7 Fix $b > 1$ , $y > 0$ , and prove that there is a unique real $x$ such that $b^x = y$ , by completing the following outline. (This is called the *logarithm of $y$ to the base $b$* .)

### 7.1 a. For any positive integer $n$ , $b^n - 1 \geq n(b - 1)$

We will prove this with induction on  $n$ . Our base case is  $n = 1$ .

$$b^1 - 1 \geq 1(b - 1)$$

$$b - 1 \geq b - 1$$

This is a true statement, so  $b^n - 1 \geq n(b - 1)$  holds for  $n = 1$ .

Now for the inductive step. For  $n > 1$ , Assume  $b^{n-1} - 1 \geq (n - 1)(b - 1)$ .

$$b^{n-1} - 1 \geq (n - 1)(b - 1)$$

We add  $b - 1$  to both sides.

$$b^{n-1} + b - 2 \geq n(b - 1)$$

Now, if we can show  $b^n - 1 \geq b^{n-1} + b - 2$ , then we can conclude  $b^n - 1 \geq b^{n-1} + b - 2 \geq n(b - 1)$ . To show  $b^n - 1 \geq b^{n-1} + b - 2$ , we can show that  $b^n - b^{n-1} \geq b - 1$ . Well,  $b^n - b^{n-1} = b^{n-1}(b - 1)$ , and since  $b^{n-1} \geq (n - 1)(b - 1) + 1$  (from the inductive hypothesis),  $b^{n-1} \geq 1$ . Thus,  $b^n - b^{n-1} \geq (b - 1)$ ,

This allows us to conclude  $b^n - 1 \geq b^{n-1} + b - 2 \geq n(b - 1)$ . We have shown  $b^n - 1 \geq n(b - 1)$  if  $b^{n-1} - 1 \geq (n - 1)(b - 1)$ .

## 7.2 b. Hence $b - 1 \geq n(b^{\frac{1}{n}} - 1)$ .

Since  $b > 1$ , it is the case that  $b^{\frac{1}{n}} > n$ , since  $(b^{\frac{1}{n}})^n = b$  would not be possible if  $0 < b^{\frac{1}{n}} < 1$ . Consider the inequality from the previous problem.

$$b^n - 1 \geq n(b - 1)$$

Since  $b^{\frac{1}{n}} > 1$ , the inequality still holds for  $b^{\frac{1}{n}}$ .

$$(b^{\frac{1}{n}})^n - 1 \geq n(b^{\frac{1}{n}} - 1)$$

$$b - 1 \geq n(b^{\frac{1}{n}} - 1)$$

## 7.3 c. If $t > 1$ and $n > \frac{b-1}{t-1}$ , then $b^{\frac{1}{n}} < t$

From the previous part, we have

$$b - 1 \geq n(b^{\frac{1}{n}} - 1)$$

We are also given  $n > \frac{b-1}{t-1}$ . Let's isolate  $t - 1$  in this inequality, since we want to show  $t > \text{something}$ .

$$t - 1 > \frac{b - 1}{n}$$

$$t > \frac{b - 1}{n} + 1$$

Since  $b > 1$ , we can note that  $n > \frac{b-1}{t-1} > 0$  as well. By inspection, we can notice that  $\frac{b-1}{n} + 1$  is present in the previous problem's inequality too.

$$\frac{b - 1}{n} \geq b^{\frac{1}{n}} - 1$$

$$\frac{b - 1}{n} + 1 \geq b^{\frac{1}{n}}$$

Thus, we have

$$t > \frac{b - 1}{n} + 1 \geq b^{\frac{1}{n}}$$

as desired.

**7.4 d. If  $w$  is such that  $b^w < y$ , then  $b^{w+\frac{1}{n}} < y$  for sufficiently large  $n$ ; to see this, apply part (c) with  $t = y \cdot b^{-w}$ .**

We take  $t = y \cdot b^{-w}$  as instructed and invoke part c.  $y > b^w$ , so  $y \cdot b^{-w} > 1$ .

$$y \cdot b^{-w} > b^{\frac{1}{n}}$$

for  $n > \frac{b-1}{y \cdot b^{-w} - 1}$ . We thus have

$$b^{w+\frac{1}{n}} < y$$

for  $n > \frac{n-1}{y \cdot b^{-w} - 1}$ .

**7.5 e. If  $b^w > y$ , then  $b^{w-\frac{1}{n}} > y$  for sufficiently large  $n$ .**

Since  $b^w > y$ , we take  $t = \frac{b^w}{y} > 1$ . Now, we once again invoke part c.

$$\frac{b^w}{y} > b^{\frac{1}{n}}$$

$$b^{w-\frac{1}{n}} > y$$

**7.6 f. Let  $A$  be the set of all  $w$  such that  $b^w < y$ , and show that  $x = \sup A$  satisfies  $b^x = y$ .**

First, it cannot be the case that  $b^x < y$ . From part d, we know there exists  $b^x < b^{x+\frac{1}{n}} < y$ . Since  $b^{x+\frac{1}{n}} < y$ ,  $x + \frac{1}{n} \in A$ , and  $x$  is thus not an upper bound.

We now also prove that  $b^x > y$  cannot hold. In such a scenario, there exists  $b^{x-\frac{1}{n}} > y$  by part e.  $b^{x-\frac{1}{n}} > y$ , so for all  $w \in A$ ,  $b^{x-\frac{1}{n}} > y > b^w$ . Thus, there is a smaller upper bound  $x - \frac{1}{n}$ , meaning,  $x$  is not the least upper bound.

Since neither  $b^x < y$  nor  $b^x > y$  can hold, it must be the case that  $b^x = y$ .

**7.7 g. Prove that this  $x$  is unique.**

Let  $y < x$ . Then,  $b^y < b^x$ , since  $b > 1$ .  $y > x$  similarly implies  $b^y > b^x$ .

**8 Prove that no order can be defined in the complex field that turns it into an ordered field.**

Since  $1 = 1^2$ , by Proposition 1.18d 1 must be positive. Similarly, because  $-1 = i^2$ ,  $-1$  is positive as well. If  $-1 > 0$  (meaning  $-1$  is positive), then by Definition 1.17a  $1+0 < 1+(-1)$  since  $1, 0, -1$  are complex numbers and  $0 < -1$ . This gives  $1 < 0$ , which is a contradiction.

**9 Suppose  $z = a + bi$  and  $w = c + di$ . Define  $z < w$  if  $a < c$ , and also if  $a = c$  but  $b < d$ . Prove that this turns the set of all complex numbers into an ordered set. Does this ordered set have the least-upper-bound property?**

We must show that if  $x, y, z$  are complex numbers, then  $x < y$  and  $y < z$  implies  $x < z$ . Assume  $x < y$  and  $y < z$  for our arbitrary  $x, y, z$ . Let  $x = x_r + x_i i$ ,  $y = y_r + y_i i$ , and  $z = z_r + z_i i$ . There are four cases:

Case 1:  $x_r < y_r$  and  $y_r < z_r$ :  
We can conclude that  $x_r < z_r$ , so  $x < z$ .

Case 2:  $x_r < y_r$  and  $y_r = z_r \wedge y_i < z_i$ :  $y_r = z_r$ ,  $z_r = y_r > x_r$ , so  $x < z$ .

Case 3:  $x_r = y_r \wedge x_i < y_i$  and  $y_r < z_r$ :  
Since  $x_r = y_r < z_r$ ,  $x < z$ .

Case 4:  $x_r = y_r \wedge x_i < y_i$  and  $y_r = z_r \wedge y_i < z_i$ :  
Since  $x_r = y_r = z_r$ , we compare the imaginary part. We have  $x_i < y_i < z_i$ , so  $x_r = z_r$  and  $x_i < z_i$ . meaning  $x < z$ .

These cases are exhaustive, as we are given  $x < y$  and  $y < z$ .

This set does not obey the least-upper-bound property. Consider the set  $S$  of all complex numbers  $a + bi$  where  $0 < a, b < 1$ . Here, the upper bounds of  $S$  must be of the form  $c + di$ , where  $c \geq 1$ . If  $c < 1$ , then there exists some  $e + fi \in S$  with  $c < e < 1$ , so then  $c + di$  would not be an upper bound.

There is clearly no least upper bound of  $S$ , as if  $c + di$  is an upper bound, then we can always find  $c + (d - 1)i$  as a smaller upper bound.

**10 Suppose that  $z = a + bi$ ,  $w = u + iv$ , and**

$$a = \left( \frac{|w| + u}{2} \right)^{\frac{1}{2}}, b = \left( \frac{|w| - u}{2} \right)^{\frac{1}{2}}$$

**Prove that  $z^2 = w$  if  $v \geq 0$  and that  $(\bar{z})^2 = w$  if  $v \leq 0$ . Conclude that every complex number (with one exception!) has two complex square roots.**

If  $v \geq 0$ , then  $z^2 =$

$$(a^2 - b^2) + 2abi$$

Here,  $a^2 = \frac{|w|+u}{2}$  and  $b^2 = \frac{|w|-u}{2}$ .

$$a^2 - b^2 = \frac{2u}{2} = u$$

$$2ab = 2\left(\frac{(|w|+u)(|w|-u)}{4}\right)^{\frac{1}{2}} = 2\left(\frac{|w|^2 - u^2}{4}\right)^{\frac{1}{2}}$$

Since  $|w|^2 = u^2 + v^2$ ,

$$2ab = 2\left(\frac{v^2}{4}\right) = v$$

since  $v \geq 0$ .

Thus,  $z^2 = u + vi = w$  if  $v \geq 0$ .

We now tackle  $v \leq 0$ , with  $\bar{z}^2 =$

$$(a^2 - b^2) - 2abi$$

Once again, the real term of  $\bar{z}^2$  is  $u$ .

For the imaginary term, we compute

$$-2ab = -2\left(\frac{|w|^2 - u^2}{4}\right)^{\frac{1}{2}} = -2\left(\frac{v^2}{4}\right) = -2\left|\frac{v}{2}\right| = -|v| = v$$

Thus,  $\bar{z}^2 = u + vi = w$  for  $v \leq 0$ .

Consider some arbitrary complex number  $w = u + vi$ .

If  $v = 0$  and  $u \neq 0$ , then it is clear there are exactly two square roots of  $w$ :  $\sqrt{u}$  and  $-\sqrt{u}$ .

If  $v \neq 0$ , then there are definitely at least two, if we take  $z$  and  $\bar{z}$ . Let there be some complex number  $z = a + bi$  that satisfies  $z^2 = w$ . It must be the case that  $a^2 - b^2 = u$  and  $2ab = v$ .

$$b = \frac{v}{2a}$$

$$a^2 - \frac{v^2}{4a^2} = u$$

$$4a^4 - 4ua^2 - v^2 = 0$$

We can use the quadratic formula to get  $a^2 = \frac{4u \pm \sqrt{16u^2 + 16v}}{8} = \frac{u \pm \sqrt{u^2 + v^2}}{2}$ . Note that only  $\frac{u + \sqrt{u^2 + v^2}}{2}$  is a possible solution, as using the minus sign would mean  $a^2 < 0$ . Clearly, if  $u \neq 0$  or  $v \neq 0$ , then  $a^2 \neq 0$ . If  $a^2 \neq 0$ , then there are two possible nonzero values for  $a$ , and since both are nonzero, both confer a valid  $b$  with  $b = \frac{v}{2a}$ . We can thus conclude that there are at most two solutions for the case where  $v \neq 0$ .

Thus, if  $v \neq 0$ , there are at most 2 solutions, of which our previous work has shown we can use  $z$  and  $\bar{z}$  as our 2 solutions. If  $v = 0$  and  $a \neq 0$ , our only solutions are  $\sqrt{u}$  and  $-\sqrt{u}$ . Lastly, for  $u, v = 0$ , there is only one solution, which is  $z = 0$ .

## 11 If $z$ is a complex number, prove that there exists an $r \geq 0$ and a complex number $w$ with $|w| = 1$ such that $z = rw$ . Are $w$ and $r$ always uniquely determined by $z$ ?

Let  $w = u + vi$  and  $z = a + bi$ . Then, for  $|w| = 1$  to be true it implies  $u^2 + v^2 = 1$ .

Thus, we want to know how many tuples  $(u, v, r)$  exist such that

$$\begin{aligned} ru &= a \\ rv &= b \\ u^2 + v^2 &= 1 \end{aligned}$$

From this we can see

$$\begin{aligned} u &= \frac{a}{r} \\ v &= \frac{b}{r} \end{aligned}$$

We then require

$$\begin{aligned} \frac{a^2}{r^2} + \frac{b^2}{r^2} &= 1 \\ a^2 + b^2 &= r^2 \end{aligned}$$

Thus, if  $r \geq 0$ , only  $r = \sqrt{a^2 + b^2}$  satisfies this. With  $r$  uniquely determined,  $w$  is too, as there is only one value of  $r^{-1}z$ .

## 12 If $z_1, \dots, z_n$ are complex, prove that

$$|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|$$

Let  $z_k = a_k + b_k i$ . Let's consider the LHS. It will be equal to

$$\sqrt{\left(\sum_{i=1}^n a_i\right)^2 + \left(\sum_{i=1}^n b_i\right)^2}$$

which equals

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^n a_i a_j + b_i b_j}$$

Let's now consider the RHS.

$$\sum_{i=1}^n \sqrt{a_i^2 + b_i^2}$$

Since both sides are nonnegative, let's square them. LHS:

$$\sum_{i=1}^n \sum_{j=1}^n a_i a_j + b_i b_j$$

RHS:

$$\begin{aligned} & \left(\sum_{i=1}^n \sqrt{a_i^2 + b_i^2}\right)^2 \\ & \sum_{i=1}^n \sum_{j=1}^n \sqrt{a_i^2 + b_i^2} \cdot \sqrt{a_j^2 + b_j^2} \\ & \sum_{i=1}^n \sum_{j=1}^m \sqrt{a_i^2 a_j^2 + b_i^2 a_j^2 + a_i^2 b_j^2 + b_i^2 b_j^2} \end{aligned}$$

Since both of these are similarly structured sums, let's try to compare the terms. We know  $a_i a_j + b_i b_j \leq |a_i a_j + b_i b_j| = \sqrt{(a_i a_j + b_i b_j)^2}$ , which equals

$$\sqrt{a_i^2 a_j^2 + 2a_i a_j b_i b_j + b_i^2 b_j^2}$$

We now want to compare  $a_i^2 a_j^2 + 2a_i a_j b_i b_j + b_i^2 b_j^2$  and  $a_i^2 a_j^2 + b_i^2 a_j^2 + a_i^2 b_j^2 + b_i^2 b_j^2$ . We notice both of these terms have  $a_i^2 a_j^2$  and  $b_i^2 b_j^2$ , so we can ignore them.

Thus we are comparing  $2a_i a_j b_i b_j$  and  $b_i^2 a_j^2 + a_i^2 b_j^2$ . Let's subtract one from the other.

$$2a_i a_j b_i b_j - b_i^2 a_j^2 - a_i^2 b_j^2 = a_i a_j b_i b_j + a_i a_j b_i b_j - b_i^2 a_j^2 - a_i^2 b_j^2$$

$$a_j b_i (a_i b_j - a_j b_i) + a_i b_j (a_j b_i - a_i b_j) \\ (a_i b_j - a_j b_i)(a_j b_i - a_i b_j) = -(a_i b_j - a_j b_i)^2$$

Since we get a negative value, it must be the case that  $b_i^2 a_j^2 + a_i^2 b_j^2 \geq 2a_i a_j b_i b_j$ . We can then conclude that

$$\sqrt{a_i^2 a_j^2 + b_i^2 a_j^2 + a_i^2 b_j^2 + b_i^2 b_j^2} \geq \sqrt{a_i^2 a_j^2 + 2a_i a_j b_i b_j + b_i^2 b_j^2}$$

Which gives

$$\sum_{i=1}^n \sum_{j=1}^m \sqrt{a_i^2 a_j^2 + b_i^2 a_j^2 + a_i^2 b_j^2 + b_i^2 b_j^2} \geq \sum_{i=1}^n \sum_{j=1}^n \sqrt{a_i^2 a_j^2 + 2a_i a_j b_i b_j + b_i^2 b_j^2}$$

So

$$\sum_{i=1}^n \sqrt{a_i^2 + b_i^2} \geq \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_i a_j + b_i b_j}$$

and finally

$$|z_1 + z_2 + \dots + z_n| \geq |z_1| + |z_2| + \dots + |z_n|$$

### 13 If $x, y$ are complex, prove that

$$||x| - |y|| \leq |x - y|$$

Let  $x = a + bi$  and let  $y = c + di$ .

Let's simplify the LHS:

$$|x| = \sqrt{a^2 + b^2}$$

$$|y| = \sqrt{c^2 + d^2}$$

$$|x| - |y| = \sqrt{a^2 + b^2} - \sqrt{c^2 + d^2}$$

$$||x| - |y|| = |\sqrt{a^2 + b^2} - \sqrt{c^2 + d^2}|$$

and now the RHS:

$$x - y = (a - c) + (b - d)i$$

$$|x - y| = \sqrt{(a - c)^2 + (b - d)^2}$$

Since both  $||x| - |y||$  and  $|x - y|$  are positive, if we square them, we preserve the inequality.

$$||x| - |y||^2 = a^2 + b^2 - 2\sqrt{(a^2 + b^2)(c^2 + d^2)} + c^2 + d^2$$

$$|x - y|^2 = a^2 - 2ac + c^2 + b^2 - 2bd + d^2$$



We notice both terms have  $a^2 + b^2 + c^2 + d^2$ , so we can ignore them.

The new LHS is:

$$-2\sqrt{a^2c^2 + b^2c^2 + a^2d^2 + b^2d^2}$$

and the new RHS:

$$-2ac - 2bd$$

We factor out  $-2$ .

LHS:

$$\sqrt{a^2c^2 + b^2c^2 + a^2d^2 + b^2d^2}$$

RHS:

$$ac + bd$$

If we square them, we will be able to compare  $\sqrt{a^2c^2 + b^2c^2 + a^2d^2 + b^2d^2}$  with  $|ac + bd|$ .

LHS:

$$a^2c^2 + b^2c^2 + a^2d^2 + b^2d^2$$

RHS:

$$a^2c^2 + 2abcd + b^2d^2$$

Both terms have  $a^2c^2 + b^2d^2$ , so we can cancel them: LHS:

$$b^2c^2 + a^2d^2$$

$$2abcd$$

We may recall from the previous problem that

$$b^2c^2 - 2abcd + a^2d^2 = (bc + ad)^2 \geq 0$$

So it must be the case that

$$b^2c^2 + a^2d^2 \geq 2abcd$$

Then, we can conclude that

$$a^2c^2 + b^2c^2 + a^2d^2 + b^2d^2 \geq a^2c^2 + 2abcd + b^2d^2$$

Thus,

$$\sqrt{a^2c^2 + b^2c^2 + a^2d^2 + b^2d^2} \geq |ac + bd| \geq ac + bd$$

Giving

$$-2\sqrt{a^2c^2 + b^2c^2 + a^2d^2 + b^2d^2} \leq -2ac - 2bd$$

Then,

$$a^2 + b^2 - 2\sqrt{(a^2 + b^2)(c^2 + d^2)} + c^2 + d^2 \leq a^2 - 2ac + c^2 + b^2 - 2bd + d^2$$

$$||x| - |y||^2 \leq |x - y|^2$$

And finally,

$$||x| - |y|| \leq |x - y|$$

**14 If  $z$  is a complex number such that  $|z| = 1$ , that is, such that  $z\bar{z} = 1$ , compute**

$$|1 + z|^2 + |1 - z|^2$$

Let  $z = a + bi$ . Let's first compute  $|1 + z|^2$ .

$$(\sqrt{(1+a)^2 + b^2})^2$$

$$a^2 + 2a + 1 + b^2$$

$$a^2 + b^2 + 2a + 1$$

Since  $a^2 + b^2 = 1$  ( $|z| = 1$ ),

$$2a + 2$$

We evaluate  $|1 - z|^2$  as well

$$(\sqrt{(1-a)^2 + b^2})^2$$

$$(1-a)^2 + b^2$$

$$1 - 2a + a^2 + b^2$$

$$2 - 2a$$

Thus, the answer is  $2a + 2 + 2 - 2a = 4$ .

**15 Under what conditions does equality hold in the Schwarz Inequality?**

The Schwarz Inequality:

$$\left| \sum_{j=1}^n a_j \bar{b}_j \right|^2 \leq \sum_{j=1}^n |a_j|^2 \sum_{j=1}^n |b_j|^2$$

Let  $a_j = c_j + d_j i$  and let  $b_j = e_j + f_j i$ .

Let's evaluate the LHS:

$$a_j \bar{b}_j = (c_j e_j + d_j f_j) + (d_j e_j - c_j f_j) i$$

$$\left| \sum_{j=1}^n (c_j e_j + d_j f_j) + (d_j e_j - c_j f_j) i \right|^2$$

$$\begin{aligned}
& \left| \sum_{j=1}^n (c_j e_j + d_j f_j) + \sum_{j=1}^n (d_j e_j - c_j f_j) i \right|^2 \\
& \left( \sqrt{\left( \sum_{j=1}^n (c_j e_j + d_j f_j) \right)^2 + \left( \sum_{j=1}^n (d_j e_j - c_j f_j) \right)^2} \right)^2 \\
& \left( \sum_{j=1}^n (c_j e_j + d_j f_j) \right)^2 + \left( \sum_{j=1}^n (d_j e_j - c_j f_j) \right)^2 \\
& \sum_{j=1}^n \sum_{k=1}^n (c_j e_j + d_j f_j)(c_k e_k + d_k f_k) + \sum_{j=1}^n \sum_{k=1}^n (d_j e_j - c_j f_j)(d_k e_k - c_k f_k) \\
& \sum_{j=1}^n \sum_{k=1}^n (c_j e_j + d_j f_j)(c_k e_k + d_k f_k) + (d_j e_j - c_j f_j)(d_k e_k - c_k f_k) \\
& \sum_{j=1}^n \sum_{k=1}^n c_j c_k e_j e_k + c_k d_j e_k f_j + c_j d_k e_j f_k + d_j d_k f_j f_k + d_j d_k e_j e_k - c_j d_k e_k f_j - c_k d_j e_j f_k + c_j c_k f_j f_k
\end{aligned}$$

On the RHS:

$$\begin{aligned}
& \sum_{j=1}^n \sum_{k=1}^n |a_j|^2 |b_k|^2 \\
& \sum_{j=1}^n \sum_{k=1}^n (c_j^2 + d_j^2)(e_k^2 + f_k^2) \\
& \sum_{j=1}^n \sum_{k=1}^n c_j^2 e_k^2 + d_j^2 e_k^2 + c_j^2 f_k^2 + d_j^2 f_k^2
\end{aligned}$$

Let's subtract them:

$$\begin{aligned}
u_{j,k} &= c_j c_k e_j e_k + c_k d_j e_k f_j + c_j d_k e_j f_k + d_j d_k f_j f_k + d_j d_k e_j e_k - c_j d_k e_k f_j - c_k d_j e_j f_k + c_j c_k f_j f_k \\
v_{j,k} &= c_j^2 e_k^2 + d_j^2 e_k^2 + c_j^2 f_k^2 + d_j^2 f_k^2 \\
0 &= \sum_{j=1}^n \sum_{k=1}^n u_{j,k} - v_{j,k}
\end{aligned}$$

For  $j = k$ ,  $u_{j,k} = v_{j,k}$ .

$$\begin{aligned}
u_{j,j} &= c_j^2 e_j^2 + c_j d_j e_j f_j + c_j d_j e_j f_j + d_j^2 f_j^2 + d_j^2 e_j^2 - c_j d_j e_j f_j - c_j d_j e_j f_j + c_j^2 f_j^2 \\
u_{j,j} &= c_j^2 e_j^2 + d_j^2 f_j^2 + d_j^2 e_j^2 + c_j^2 f_j^2 = v_{j,j}
\end{aligned}$$

Thus, we only consider when  $j \neq k$ . We can try to merge  $u_{j,k} - v_{j,k}$  with  $u_{k,j} - v_{k,j}$ . Note that  $u_{j,k} = u_{k,j}$ .

$$\sum_{j=1}^n \sum_{k=j+1}^n 2u_{j,k} - v_{j,k} - v_{k,j}$$

We can pair  $c_j^2 e_k^2$  with  $c_k^2 e_j^2$  and  $2c_j c_k e_j e_k$ , and so on.

$$\sum_{j=1}^n \sum_{k=j+1}^n r_{j,k} - (c_j e_k - c_k e_j)^2 - (d_j f_k - d_k f_j)^2 - (c_j f_k - c_k f_j)^2 - (d_j e_k - d_k e_j)^2$$

Here  $r_{j,k}$  is what we couldn't group up.

$$r_{j,k} = 2c_k d_j e_k f_j + 2c_j d_k e_j f_k - 2c_j d_k e_k f_j - 2c_k d_j e_j f_k$$

$$r_{j,k} = 2(e_k f_j (c_k d_j - c_j d_k) + e_j f_k (c_j d_k - c_k d_j))$$

$$r_{j,k} = 2(e_k f_j - e_j f_k)(c_k d_j - c_j d_k)$$

We can try to create perfect squares again, such as  $((c_j f_k - c_k f_j) + (d_j e_k - d_k e_j))^2$

$$\begin{aligned} & ((c_j f_k - c_k f_j) + (d_j e_k - d_k e_j))^2 = \\ & (c_j f_k - c_k f_j)^2 + 2c_j d_j e_k f_k - 2c_k d_j e_k f_j - 2c_j d_k e_j f_k + 2c_k d_k e_j f_j + (d_j e_k - d_k e_j)^2 \end{aligned}$$

Similarly,

$$\begin{aligned} & ((c_k e_j - c_j e_k) + (d_j f_k - d_k f_j))^2 = \\ & (c_k e_j - c_j e_k)^2 + 2c_k d_j e_j f_k - 2c_j d_j e_k f_k - 2c_k d_k e_j f_j + 2c_j d_k e_k f_j + (d_j f_k - d_k f_j)^2 \end{aligned}$$

Note that adding these two terms together cancels out the  $2c_k d_k e_j f_j$  and  $2c_j d_j e_k f_k$  terms.

$$\begin{aligned} & ((c_j f_k - c_k f_j) + (d_j e_k - d_k e_j))^2 + ((c_k e_j - c_j e_k) + (d_j f_k - d_k f_j))^2 = \\ & (c_j f_k - c_k f_j)^2 + (d_j e_k - d_k e_j)^2 + (c_k e_j - c_j e_k)^2 + (d_j f_k - d_k f_j)^2 + \\ & -2c_k d_j e_k f_j - 2c_j d_k e_j f_k + 2c_k d_j e_k f_j + 2c_k d_j e_j f_k \end{aligned}$$

Note that this is exactly the term we have in the summation, but negated!

$$(c_k e_j - c_j e_k)^2 = (c_j e_k - c_k e_j)^2$$

$$2u_{j,k} - v_{j,k} - v_{k,j} = -((c_j f_k - c_k f_j) + (d_j e_k - d_k e_j))^2 - ((c_k e_j - c_k e_j) + (d_j f_k - d_k f_j))^2$$

$$\sum_{j=1}^n \sum_{k=j+1}^n -((c_j f_k - c_k f_j) + (d_j e_k - d_k e_j))^2 - ((c_k e_j - c_k e_j) + (d_j f_k - d_k f_j))^2$$

We require that this summation equal to 0. However, we can see all terms are squares. Thus, all of these squares must be  $0^2$ .

We have:

$$c_j f_k - c_k f_j + d_j e_k - d_k e_j = 0$$

and

$$c_k e_j - c_k e_j + d_j f_k - d_k f_j = 0$$

The first equation gives

$$c_j f_k + d_j e_k = c_k f_j + d_k e_j$$

The second gives

$$c_k e_j - d_k f_j = c_k e_j - d_j f_k$$

We can see that  $c_j f_k + d_j e_k$  is simply just  $Im(a_j b_k)$  and  $c_k f_j + d_k e_j$  is  $Im(a_k b_j)$ . Similarly,  $c_k e_j - d_k f_j$  is  $Re(a_k b_j)$  and  $c_k e_j - d_j f_k$  is  $Re(a_j b_k)$ .

Since we have  $Im(a_j b_k) = Im(a_k b_j)$  and  $Re(a_k b_j) = Re(a_j b_k)$ , It must be the case that  $a_k b_j = a_j b_k$  for all  $j, k$ .

Assume  $b_j \neq 0$  for all  $j$ . Then,

$$\frac{a_j}{b_j} = \frac{a_k}{b_k}$$

for all  $j, k$ . This means that for all  $j$ ,  $a_j = cb_j$ , for some constant complex number  $c$ .

Now assume there exists some  $b_j = 0$ . For all  $k$ , we have

$$a_k b_j = a_j b_k = 0$$

This means  $a_j b_k = 0$  for all  $k$ . Thus, we can conclude that for all  $k$   $b_k = 0$  or  $a_j = 0$ .

If  $a_j = 0$ , we still have that for all  $j$ ,  $a_j = cb_j$ , since  $0 = c \cdot 0$ .  $b_k = 0$  for all  $k$  is a special case.

We can finally conclude that the Schwarz Inequality is an equality when for some constant  $c$  and all  $j$ ,  $a_j = cb_j$ , or  $b_j = 0$  for all  $j$ .

**16 Suppose  $k \geq 3$ ,  $x, y \in \mathbb{R}^k$ ,  $|x - y| = d > 0$ , and  $r > 0$ . Prove:**

**16.1 a. If  $2r > d$ , there are infinitely many  $z \in \mathbb{R}^k$  such that  $|z - x| = |z - y| = r$ .**

We are given that  $|x - y| = d$  and  $|z - x| = |z - y| = r$ . Let's write the latter out as a summation, as we would like to know under what conditions does  $z$

satisfy the property that it is equidistant from  $x$  and  $y$ .

$$\begin{aligned}
\sum_{i=1}^k (z_i - x_i)^2 &= \sum_{i=1}^k (z_i - y_i)^2 \\
\sum_{i=1}^k (z_i - x_i)^2 - (z_i - y_i)^2 &= 0 \\
\sum_{i=1}^k z_i^2 - 2x_i z_i + x_i^2 - z_i^2 + 2y_i z_i - y_i^2 &= 0 \\
\sum_{i=1}^k -2x_i z_i + 2y_i z_i + x_i^2 - y_i^2 &= 0 \\
\sum_{i=1}^k 2z_i(y_i - x_i) + (x_i + y_i)(x_i - y_i) &= 0 \\
\sum_{i=1}^k (x_i + y_i - 2z_i)(x_i - y_i) &= 0
\end{aligned}$$

We can write this as a dot product.

$$(x + y - 2z) \cdot (x - y) = 0$$

Intuitively (Consider the 2D or 3D case), we know that  $z$  is a vector which takes the form  $m + v$ , where  $m = \frac{x+y}{2}$ . With this substitution, we get a new dot product:

$$-2v \cdot (x - y) = 0$$

We now turn it back into a summation:

$$\sum_{i=1}^k v_i \cdot 2(y_i - x_i) = 0$$

We would like to show that there are an infinite number of  $v$  such that  $|v| = 1$ . This is because the other condition we must satisfy is:

$$|z - x| = r$$

$$|m + v - x| = r$$

$$\sum_{i=1}^k (m_i + v_i - x_i)^2 = r^2$$

$$\sum_{i=1}^k m_i^2 + 2m_i v_i - 2m_i x_i + v_i^2 - 2v_i x_i + x_i^2 = r^2$$

We can pull out the  $\sum_{i=1}^k v_i^2$  as well as  $\sum_{i=1}^k (m_i - x_i)^2$

$$|v|^2 + |m - x|^2 + \sum_{i=1}^k 2m_i v_i - 2v_i x_i = r^2$$

$$|v|^2 + |m - x|^2 + \sum_{i=1}^k v_i (2m_i - 2x_i) = r^2$$

Recall that  $m = \frac{x+y}{2}$ .

$$|v|^2 + |m - x|^2 + \sum_{i=1}^k v_i (x_i + y_i - 2x_i) = r^2$$

$$|v|^2 + |m - x|^2 + \sum_{i=1}^k v_i (y_i - x_i) = r^2$$

We already know that  $\sum_{i=1}^k v_i \cdot 2(y_i - x_i) = 0$ . But that is just  $2 \sum_{i=1}^k v_i (y_i - x_i)$ , so  $\sum_{i=1}^k v_i (y_i - x_i) = 0$ , too. Thus,

$$|v|^2 + |m - x|^2 = r^2$$

$$|v|^2 = r^2 - |m - x|^2$$

Is this satisfiable? Well, let's compute  $|m - x|^2$ .

$$\sum_{i=1}^k \left( \frac{x_i + y_i}{2} - x_i \right)^2$$

$$\sum_{i=1}^k \left( \frac{y_i - x_i}{2} \right)^2$$

$$\frac{1}{4} \sum_{i=1}^k (y_i - x_i)^2$$

$\sum_{i=1}^k (y_i - x_i)^2 = |x - y|^2 = d^2$ , so  $|m - x|^2 = \frac{1}{4}d^2$ . We can thus conclude that  $|m - x| = \frac{1}{2}d$ . Since  $r > \frac{1}{2}d$ ,  $r^2 - |m - x|^2$  is guaranteed to be positive.

We return to trying to prove that there are infinitely many  $v$  that satisfy  $-2v \cdot (x - y) = 0$  and  $|v| = 1$ . If this were true, then there would exist  $\alpha$  such that  $\alpha|v|^2 = r^2 - |m - x|^2$  since  $r^2 - |m - x|^2$  is positive, and thus there would be an infinite number of  $v'$  satisfying  $|v'|^2 = r^2 - |m - x|^2$  since for every unique  $v$  there exists  $v' = \sqrt{\alpha}v$  satisfying both  $\alpha v \cdot (y - x) = 0$  and  $|\sqrt{\alpha}v|^2 = r^2 - |m - x|^2$ . Thus, an infinite number of  $z = m + v'$  would exist too.

Now, since we have  $k \geq 3$ , we should abuse this fact. We will perform case-work.

Our first case is that there exists distinct integers  $1 \leq i, j \leq k$  such that  $y_i - x_i \neq 0$  and  $y_j - x_j \neq 0$ . We pick another arbitrary integer  $l$  such that  $1 \leq l \leq k$  and  $i \neq l, j \neq l$ . For the remaining components of  $v$ , we let them equal 0. We must satisfy the following conditions:

$$v_i(y_i - x_i) + v_j(y_j - x_j) + v_l(y_l - x_l) = 0$$

$$v_i^2 + v_j^2 + v_l^2 = 1$$

Let  $0 < |v_l| < 1$ . Then, our conditions become

$$v_i(y_i - x_i) + v_j(y_j - x_j) = -v_l(y_l - x_l)$$

$$v_i^2 + v_j^2 = 1 - v_l^2$$

Let's substitute the right hand sides with  $c, d$ , where  $0 < d < 1$ . Let's also substitute  $y_i - x_i$  with  $a$  and  $y_j - x_j$  with  $b$ .

$$v_i = \frac{c - bv_j}{a}$$

$$v_i^2 + v_j^2 = d$$

$$\frac{c^2 - 2bcv_j + b^2v_j^2}{a^2} + v_j^2 = d$$

$$c^2 - 2bcv_j + b^2v_j^2 + a^2v_j^2 = a^2d$$

$$(a^2 + b^2)v_j^2 - 2bcv_j + c^2 - a^2d = 0$$

$$v_j = \frac{2bc \pm \sqrt{4b^2c^2 - 4(a^2 + b^2)(c^2 - a^2d)}}{2a^2 + 2b^2}$$

$$v_j = \frac{2bc \pm \sqrt{-4a^2c^2 + 4a^4d + 4a^2b^2d}}{2a^2 + 2b^2}$$

We can see that  $v_j$  exists when  $c \geq 0$ . Of course, there are an infinite number of  $v_l$  we can choose such that  $-v_l(y_l - x_l) > 0$ . Thus, there are an infinite number of solutions to the required conditions.

In our second case, there is only one  $1 \leq i \leq k$  such that  $y_i - x_i \neq 0$ , and all other  $y_j - x_j = 0$  for  $j \neq i$ . Note that  $i$  must exist, as  $d > 0$ , so  $x \neq y$ . Thus, we let  $j, l$  be integers such that  $1 \leq j, l \leq l$  and  $i \neq j$  and  $i \neq l$  and  $j \neq l$ , with  $y_j - x_j = 0$  and  $y_l - x_l = 0$ .

In this case, the conditions we must satisfy are:

$$v_i(y_i - x_i) = 0$$



$$v_i^2 + v_j^2 + v_l^2 = 1$$

Here, we must have  $v_i = 0$ . However, for each  $0 < |v_j| < 1$ , there exist 2 values of  $v_l$  satisfying  $v_j^2 + v_l^2 = 1$ . Thus, because there exist an infinite amount of real numbers between 0 and 1, there are an infinite number of  $v$  satisfying our conditions, and an infinite number of  $z$  too as a result.

### 16.2 b. If $2r = d$ , there is exactly one such $z$

Recall that  $|v|^2 + |x - m|^2 = r^2$  must hold. In this case, since  $r = \frac{d}{2}$ ,  $|v|^2 + \frac{d^2}{4} = \frac{d^2}{4}$ . Clearly, the only option is  $v = 0$ , so there is only  $z = m$  satisfying our conditions.

### 16.3 c. If $2r < d$ , there is no such $z$

By similar logic, if  $r^2 < \frac{d^2}{4}$ , then  $|v|^2 < 0$ , which cannot be possible, so there are no such  $z$ .

If  $k = 2$ , then there are 2 such  $z$  for  $2r > d$ , 1 for  $2r = d$ , and 0 otherwise. 2  $z$  exist for  $2r > d$ , as we have  $v_1(y_1 - x_1) + v_2(y_2 - x_2) = 0$  and  $v_1^2 + v_2^2 = 1$ , so if  $y_1 - x_1 \neq 0$  then the resulting quadratic equation on  $v_2$  has two solutions:

$$a^2 v_2^2 + v_2^2 = 1$$

$$v_2^2 = \frac{1}{a^2 + 1}$$

For  $k = 1$ , there are no such  $z$  for  $2r > d$ . Consider the equation

$$|z - x| = |z - y|$$

This means either  $z - x = z - y$  or  $z - x = y - z$ . In the former case, we get  $x = y$ , which is not true. In the latter, we get  $z = \frac{x+y}{2}$ . Thus, we have one solution for  $2r = d$ , and no  $z$  otherwise.

## 17 Prove that

$$|x + y|^2 + |x - y|^2 = 2|x|^2 + 2|y|^2$$

**if  $x \in \mathbb{R}^k$  and  $y \in \mathbb{R}^k$ . Interpret this geometrically, as a statement about parallelograms.**

Let's expand the left hand side.

$$|x + y|^2 + |x - y|^2$$

$$\begin{aligned}
& \sum_{i=1}^k (x_i + y_i)^2 + \sum_{i=1}^k (x_i - y_i)^2 \\
& \sum_{i=1}^k x_i^2 + 2x_i y_i + y_i^2 + x_i^2 - 2x_i y_i + y_i^2 \\
& \sum_{i=1}^k 2x_i^2 + 2y_i^2 \\
& 2 \sum_{i=1}^k x_i^2 + 2 \sum_{i=1}^k y_i^2 \\
& 2|x|^2 + 2|y|^2
\end{aligned}$$

We can interpret this as the sum of the squared lengths of the diagonals of a parallelogram equals twice the sum of the squared side lengths.

**18 If  $k \geq 2$  and  $x \in \mathbb{R}^k$ , prove that there exists  $y \in \mathbb{R}^k$  such that  $y \neq 0$  but  $x \cdot y = 0$ . Is this also true if  $k = 1$ ?**

There are two cases. The first case is if there exists  $1 \leq i \leq k$  such that  $x_i = 0$ . In this case, we let  $y_i = 1$ , and  $y_j = 0$  for  $j \neq i$ . In this case  $x \cdot y = 0$  but  $y \neq 0$ .

Otherwise, there must exist  $1 \leq i, j \leq k$  such that  $x_i \neq 0$  and  $x_j \neq 0$ . In that case, we can let  $y_l = 0$  for  $l \neq i, l \neq j$ , and we must then solve the following equation:

$$x_i y_i + x_j y_j = 0$$

Solving for  $y_i$ ,

$$y_i = \frac{-x_j y_j}{x_i}$$

Since  $x_i \neq 0$ ,  $-\frac{x_j}{x_i}$  is indeed a real number. Now, for any value of  $y_j$ , We take  $y_j = 1$  and  $y_i = -\frac{x_j}{x_i}$ , showing indeed there exists  $y \neq 0$  (since  $y_j = 1$ ) such that  $x \cdot y = 0$ .

If  $k = 1$ , this is not true. We want:

$$xy = 0$$

with  $y \neq 0$ . This means  $x = 0$ , so it is not true for all  $x \in \mathbb{R}$ .

**19 Suppose  $a \in \mathbb{R}^k, b \in \mathbb{R}^k$ . Find  $c \in \mathbb{R}^k$  and  $r > 0$  such that**

$$|x - a| = 2|x - b|$$

**if and only if  $|x - c| = r$ .**

If  $|x - a| = 2|x - b|$ , then we can square this expression to conclude

$$\sum (x_i - a_i)^2 = 4 \sum (x_i - b_i)^2$$

$$\sum (x_i - a_i)^2 - 4 \sum (x_i - b_i)^2 = 0$$

$$\sum x_i^2 - 2a_i x_i + a_i^2 - 4x_i^2 + 8b_i x_i - 4b_i^2 = 0$$

Our goal is to transform this into the form  $|x - c| = r$ , so we want to make a perfect square on the left hand side of the form  $(x_i - c_i)^2$ .

$$\sum -3x_i^2 + (8b_i - 2a_i)x_i + a_i^2 - 4b_i^2 = 0$$

$$\sum -3x_i^2 + 2(4b_i - a_i)x_i + a_i^2 - 4b_i^2 = 0$$

$$\sum x_i^2 - 2\left(\frac{4}{3}b_i - \frac{1}{3}a_i\right)x_i - \frac{a_i^2}{3} + \frac{4b_i^2}{3} = 0$$

We complete the square:

$$\sum x_i^2 - 2\left(\frac{4}{3}b_i - \frac{1}{3}a_i\right)x_i + \left(\frac{4}{3}b_i - \frac{1}{3}a_i\right)^2 = \left(\frac{4}{3}b_i - \frac{1}{3}a_i\right)^2 + \frac{a_i^2}{3} - \frac{4b_i^2}{3}$$

$$\sum \left(x_i - \left(\frac{4}{3}b_i - \frac{1}{3}a_i\right)\right)^2 = \frac{16}{9}b_i^2 - \frac{8}{9}a_i b_i + \frac{1}{9}a_i^2 + \frac{a_i^2}{3} - \frac{4b_i^2}{3}$$

$$\left|x - \left(\frac{4}{3}b - \frac{1}{3}a\right)\right|^2 = \frac{4}{9}a^2 - \frac{8}{9}a_i b_i + \frac{4}{9}b_i^2$$

$$\left|x - \left(\frac{4}{3}b - \frac{1}{3}a\right)\right|^2 = \frac{4}{9} \sum a_i^2 - 2a_i b_i + b_i^2 = \frac{4}{9} \sum (a_i - b_i)^2 = \frac{4}{9} |a - b|^2$$

We can thus conclude that  $c = \frac{4}{3}b - \frac{1}{3}a$  and  $r^2 = \frac{4}{9}|a - b|^2$ . Thus,  $3c = 4b - a$  and  $3r = 2|b - a|$  for  $|x - a| = 2|x - b|$ .

**20** With reference to the Appendix, suppose that property III were omitted from the definition of a cut. Keep the same definition of order and addition. Show that the resulting set has the least-upper-bound property, that addition satisfies axioms (A1) to (A4) (with a slightly different zero element!) but that (A5) fails.

### 20.1 Least-upper-bound property

Let  $A$  be a bounded subset of  $R$ . Our proof is identical to the original: Let  $\gamma$  be union of  $A$ .  $\gamma$  satisfies Property I, because if  $A$  is bounded then there exists  $\beta$  which is an upper bound of  $A$ .  $\gamma \subset \beta$ , and because  $\beta$  is bounded (by Property I) so is  $\gamma$ .  $\gamma$  also satisfies property II. Let  $r < s$  and let  $s \in \gamma$ . This means  $s \in \alpha$  for some  $\alpha \in A$ . Since  $\alpha \in R$ ,  $\alpha$  obeys property II, so  $r \in \alpha$  because  $r < s$ . Thus,  $r \in \gamma$  too.

Now we show that  $\gamma = \sup A$ .  $\gamma$  is clearly an upper bound of  $A$ , since if  $\alpha \in A$ , then  $\alpha \leq \gamma$  because all  $r \in \alpha$  satisfy  $r \in \gamma$ . Now let  $\gamma_2$  be an upper bound of  $A$  with  $\gamma_2 < \gamma$ . That means there exists  $r \in \gamma$  such that  $r \notin \gamma_2$ .

**Lemma 20.1** *If  $\alpha \in R$  and  $r \notin \alpha$ , then  $r$  is an upper bound of  $\alpha$ .*

Proof: Assume  $r$  is not an upper bound of  $\alpha \in R$ . That means there exists  $r_2 \in \alpha$  such that  $r < r_2$ . However, this violates property II, which is a contradiction.

Since  $r \in \gamma$ , there exists  $\alpha \in A$  such that  $r \in \alpha$ . Thus,  $\gamma_2$  cannot be an upper bound of  $A$ , since  $\alpha \in A$  and  $r \in \alpha$  while  $r \notin \gamma_2$ . By our lemma  $r$  is an upper bound of  $\gamma_2$ , so for all  $s \in \gamma_2$ ,  $s \leq r$ , and since  $r \in \alpha$ ,  $s \in \alpha$  by Property II. Thus,  $\gamma_2 \subset \alpha$ , and since  $r \notin \gamma_2$ ,  $\gamma_2$  is a proper subset of  $A$ , meaning  $\gamma_2 < \alpha$ .

We can see that  $\gamma$  must be the supremum of  $A$ , since no  $\gamma_2 < \gamma$  is an upper bound of  $A$ .

### 20.2 A1

Let  $\alpha, \beta \in R$ .

We want to show  $\alpha + \beta \in R$ .

First we must show that  $\alpha + \beta \neq Q$ .

Let  $a \in Q$  be an upper bound on  $\alpha$ .  $a$  must exist, as since Property I states  $\alpha \neq Q$ , there exists  $r \in Q$  such that  $r \notin \alpha$ . Since  $r \notin \alpha$ ,  $r$  is an upper bound of  $\alpha$  by our lemma. We define  $b$  similarly as an upper bound of  $\beta$ .

Now, for all  $x + y$  such that  $x \in \alpha$  and  $y \in \beta$  (and thus,  $x + y \in \alpha + \beta$ ),  $x + y < a + b$  since  $x < a$  and  $y < b$ . We can conclude that  $a + b \notin \alpha + \beta$  as a result. Thus,  $\alpha + \beta \neq Q$  since  $a + b \in Q$  and  $a + b \notin \alpha + \beta$ .

Second, we show that if  $r \in \alpha + \beta$  and  $s < r$ , then  $s \in \alpha + \beta$ . Since  $r \in \alpha + \beta$ ,  $r = x + y$  for some  $x \in \alpha$  and  $y \in \beta$ . Since  $s < x + y$ , we  $x + y - s > 0$ . Since  $x + y - s > 0$ ,  $s - x - y < 0$ . Thus,  $x + (s - x - y) < x$  since  $(s - x - y) < 0$ , giving  $s - y < x$ . And since  $x \in \alpha$  and  $s - y < x$ ,  $s - y \in \alpha$  too. Thus, we can have  $s - y \in \alpha$  and  $y \in \beta$  such that  $s - y + y \in \alpha + \beta$ , so  $s \in \alpha + \beta$ .

### 20.3 A2

Let  $\alpha, \beta \in R$ .

We would like to show  $\alpha + \beta = \beta + \alpha$ . If  $x + y \in \alpha + \beta$  such that  $x \in \alpha$  and  $y \in \beta$ , then  $y + x \in \beta + \alpha$  since  $y \in \beta$  and  $x \in \alpha$ . The sets must be the same.

### 20.4 A3

Let  $\alpha, \beta, \gamma \in R$ .

We would like to show  $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$ .

Let  $x + y$  be an element of  $(\alpha + \beta) + \gamma$  such that  $x \in \alpha + \beta$  and  $y \in \gamma$ . Since  $x \in \alpha + \beta$ , there exist  $x_1 \in \alpha$  and  $x_2 \in \beta$  such that  $x_1 + x_2 = x$ . Thus,  $x + y \in \alpha + (\beta + \gamma)$ , since  $x_1 \in \alpha$  and  $x_2 + y \in \beta + \gamma$  because  $x_2 \in \beta$  and  $y \in \gamma$ .

### 20.5 A4

We want to show  $R$  contains an element  $0$  such that  $0 + \alpha = \alpha$  for every  $\alpha \in R$ .

We define  $0$  to be the set of all rationals less than or equal to  $0$ .

Let  $\alpha \in R$ . Let  $x + y \in 0 + \alpha$ , where  $x \leq 0$  and  $y \in \alpha$ .  $x + y \in \alpha$  must hold, as  $x + y \leq y$  since  $x \leq 0$ , and  $y \in \alpha$ . Thus, by Property II,  $x + y \in \alpha$ . Now let  $x \in \alpha$ . Then,  $x \in 0 + \alpha$  too since we can take  $0 \in 0$  as well as  $x \in \alpha$  to get  $0 + x = x \in 0 + \alpha$ .

## 20.6 A5

Let  $\alpha \in R$  be the set of all rationals less than 0. We will show there does not exist  $\beta \in R$  such that  $\alpha + \beta = 0$ .

The first case is if  $\beta \leq 0$ . In this case, all  $x + y \in \alpha + \beta$  such that  $x \in \alpha$  and  $y \in \beta$  are less than 0, since  $x < 0$  and  $y \leq 0$ .  $y \leq 0$  because  $\beta \leq 0$ . If  $y > 0$ , then  $\beta \leq 0$  would be a contradiction as  $y \in \beta$  but  $y \notin 0$ . So, it must be the case that  $\beta > 0$ .

Now we consider  $\beta > 0$ . Let  $y \in \beta$  be some rational such that  $y > 0$ .  $y$  must exist, as  $\beta > 0$ . Thus, there is some  $y \in \beta$  such that  $y \notin 0$ , meaning  $y > 0$ . Since  $\alpha$  consists of all rationals less than 0, there exists a rational  $x$  such that  $-y < x < 0$ . since  $-y < x$ ,  $x + y > 0$ , meaning  $x \in \alpha$  and  $y \in \beta$ , so  $x + y \in \alpha + \beta$ . But because  $x + y > 0$ ,  $x + y \notin 0$ , so  $\alpha + \beta$  cannot be 0.